Successful Pregnancy Outcome in a Patient Treated with Pembrolizumab and Exposed to Fluoro-Deoxyglucose PET/CT

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2. Case Report

A multiparous patient, diagnosed with Hodgkin's lymphoma (HL), nodular sclerosis subtype, in October 2020, was already undergoing treatment for type 1 diabetes with Insulin Lispro and Glargine. She underwent chemotherapy treatment according to the ABVD protocol (Doxorubicin, Bleomycin, Vinblastine, and Dacarbazine) from November 2020 to April 2021, with biweekly infusion cycles. Due to persistent disease after this regimen, second-line therapy with Dexamethasone, Cisplatin, and Cytarabine was initiated. The patient refused to undergo an autologous transplant. Subsequently, given disease progression, she underwent chemotherapy with Brentuximab and Bendamustine, followed by Brentuximab Vedotin (four cycles between January 2021 and June 2022). However, due to limited clinical response, treatment with pembrolizumab was started (11 cycles between September 2022 and May 2023). Following a positive pregnancy test performed due to prolonged amenorrhea, the patient was referred to our maternal–fetal medicine center. The last menstrual period was unknown, and an ultrasound dating the pregnancy performed on 6 June 2023, revealed a gestational age of approximately 24 weeks, estimating the last menstrual period (LMP) to be 20 December 2022. After determining the start of the gestational period, we reconstructed the treatments and diagnostic tests performed during the pregnancy (Figure 1):

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Treatment with pembrolizumab from July 2022 to December 2022 (six cycles of 200 mg) with a possible sixth cycle occurring in the periconceptional period.

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Treatment with pembrolizumab, 200 mg in February, March, April, May during the first and second trimester.

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18F-FDG PET/CT on 18 January 2023, first trimester.

Figure 1.

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Timeline sequence. This timeline illustrates the sequence of events from October 2020 to June 2023, documenting the diagnosis, treatment milestones, and key personal developments of our patient. It highlights pivotal moments such as the diagnosis of Classical Hodgkin lymphoma, pembrolizumab treatments, FDG-PET examination during pregnancy, and the pregnancy diagnosis at 24 weeks of estimated gestational age.

After a detailed evaluation of the fetal anatomy according to the SIEOG guidelines for the anomaly scan [13], excluding major malformations, a multidisciplinary counseling was immediately offered to the patient involving hematologists and neonatologists to detect the best maternal therapeutic and diagnostic strategy to ensure the best fetal–neonatal outcomes. The exclusion of major malformations, adequate fetal growth, and a functional fetal cardiac and maternal–fetal velocimetric evaluation, at the time of the examination (24 weeks), did not allow us to exclude the possible appearance of developmental malformations and/or appearance of placental insufficiency and fetal growth restriction later during the pregnancy; few studies on the effects of pembrolizumab in pregnancy allow us to exclude postnatal neurocognitive pathologies.

Expressing the desire to continue the pregnancy, the patient underwent ultrasound scans regularly to monitor fetal growth and to exclude any structural malformative pathologies. In addition, by scanning the lymph node stations, the evaluation of maternal clinical conditions was monitored. The timing and mode of delivery were discussed with hematologists, assessing the absence of the need to carry out the delivery before the 37th week of gestation, except in the case of maternal and/or fetal complications. In September 2023, at 37.5 weeks of gestation, the patient was hospitalized with a diagnosis of premature rupture of membranes (PROM). On the same day, she delivered a male newborn weighing 2650 g, with an APGAR score of 8 and 9 at 1 and 5 min, respectively.

The patient and newborn were discharged on the third day of the puerperium in good health. The newborn underwent a routine follow-up in accordance with the SIN guidelines for the first six months (Italian Society of Neonatology) [14]. No cognitive or neurobehavioral deficits were found during neonatal checks.